Relational Calculus (Logic)

{p.prod_name |
Product(p) ∧
p.prod_price > 20 ∧
p.prod_manufacturer = ‘GizmoWorks’}

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

$$\{p.prod_price \times 0.0825 \mid \exists m(Product(p) \wedge Manufacturer(m) \wedge p.prod_manufacturer = m.man_name \wedge m.man_city = \text{‘Houston’} \wedge m.man_state = \text{‘TX’})\}$$

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Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

$$\{m.man_name \mid Manufacturer(m) \wedge \neg \exists p(Product(p) \wedge p.prod_manufacturer = m.man_name)\}$$

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Notational Details

- Set-former notation: $\{x.\, a, x.\, b, \ldots \mid \varphi\}$

- $Product(p)$ also written as $p \in Product$.

Quantification

$$\{p.prod_price \times 0.0825 \mid \exists m(Product(p) \wedge Manufacturer(m) \wedge p.prod_manufacturer = m.man_name \wedge m.man_city = \text{‘Houston’} \wedge m.man_state = \text{‘TX’})\}$$

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